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ORIGINAL ARTICLE

School-Based Interventions Targeting Challenging Behavior of Adolescents with Developmental Disabilities: A Meta-Analysis

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Abstract

Challenging behavior tends to increase in levels during adolescence for individuals with developmental disabilities (DD). If not addressed, this can lead to negative post-school outcomes for adolescents with DD. The purposes of this meta-analysis were to examine the effects of behavioral intervention in reducing challenging behaviors of adolescents with DD and to identify variables that could potentially moderate these effects. This meta-analysis included thirty single-case design studies that met standards for methodological rigor and experimental control. Overall, behavioral interventions were found to yield moderate effects for adolescents with DD (overall Tau-U = 0.68, *SD* = 0.04, [0.62, 0.73]) across settings and types of challenging behaviors. With regards to participant and setting characteristics, verbal ability and classroom setting were found to moderate the effects of the intervention outcomes with statistical significance. No other variables produced similar findings. Furthermore, statistically significant differences were found between interventions that included or did not include planned reinforcement as an intervention component. Several implications for research and practice are discussed.

Keywords Adolescent · Challenging behavior · School · Intervention · Meta-analysis

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Individuals with developmental disabilities (DD) often engage in challenging behaviors, such as aggression (Brosnan & Healy, 2011), self-injury (Gregori et al., 2018), or stereotypy (Lanovaz & Sladeczek, 2012). Challenging behaviors are behaviors that impede learning, threaten the physical and health safety of the individual and others, and can lead to isolation from the community (Emerson et al., 2001; Matson et al., 2010). Recent estimates indicate that the prevalence of challenging behavior is as high as 60% for individuals with intellectual disability (ID), 94% for individuals with autism spectrum disorder (ASD), and 93.7% for individuals with comorbid ASD and ID (Simó-Pinatella et al., 2019). Furthermore, challenging behavior is likely to persist over time (Murphy et al., 2005) with rates reaching their peak during adolescence (Billstedt et al., 2007; Shattuck et al., 2007).

While students in all age groups can engage in challenging behavior, engaging in challenging behavior is of particular concern during adolescence. Generally, challenging behavior in older students is associated with delinquency, elevated risk of substance use, potential involvement in the juvenile justice system, and other high-risk behaviors (Kuhn et al., 2015). Furthermore, as students get older, challenging behavior is more likely to be associated with academic underachievement and truancy (Fergusson & Horwood, 1995). Although for adolescents with DD, research has also shown that the presence of challenging behavior is a significant predictor of negative post-school outcomes (Wehman et al., 2015) and a lower quality of life (Biggs & Carter, 2016).

Biggs and Carter (2016) surveyed 389 parents and caregivers about the perceived quality of life of adolescents with ASD and ID. The results of their study indicated that adolescents with ASD and ID were perceived to have a lower quality of life within the school environment if they engaged in challenging behavior. Wehman et al. (2015) examined the variables associated with successful post-high school employment outcomes for individuals with disabilities and identified positive classroom behavior as a predictor of higher rates of post-school employment. The chronicity of challenging behavior and the findings of the transition literature suggested the need to address challenging behavior of adolescents with DD to promote positive post-school outcomes (Biggs & Carter, 2016; Wehman et al., 2015). Fortunately, the challenging behavior literature has shown that with intervention, individuals with DD can experience improvements in challenging behavior and other prosocial behaviors, which can lead to positive postsecondary outcomes.

For example, several reviews have identified behavioral interventions, such as functional communication training (FCT) and differential reinforcement of other behaviors (DRO), as evidence-based practices for individuals with DD (Gerow et al., 2018; Gregori et al., 2019) and have been shown to produce a large effect in decreasing challenging behavior (Chezan et al., 2018). However, much of the current research on interventions to reduce challenging behavior has focused on preschool or elementary-aged students with ASD (Severini et al., 2018). Reviews focusing on challenging behavior of older students with varying DD are rather limited. To our knowledge, there are only two reviews that have synthesized the available literature on school-based challenging behavior interventions for adolescents with DD (David et al., 2021; Rivera et al., 2019).

Rivera et al. (2019) examined the quality and strength of evidence of school-based challenging behavior interventions for adolescents with ASD using What Works Clearinghouse (WWC) quality and evidence standards (WWC, 2020). Of the 40 articles included in the review, just over half had moderate or strong evidence of effectiveness. Although this review provided initial evidence for the efficacy of school-based challenging behavior interventions, its scope was limited. Only students with ASD were included in the review, which limits the generalizability of the findings to individuals with other DD. Additionally, because intervention efficacy was determined using visual analysis, conclusions could not be drawn regarding variables that could potentially moderate the overall intervention effectiveness.

David et al. (2021) examined the quality of the available literature on school-based challenging behavior interventions for adolescents with DD using the standards developed by the Council for Exceptional Children (CEC, 2014). Overall, 53 studies were initially included in the review, but only 32 studies were considered high quality. Findings from the review indicated that antecedent-based interventions were used most frequently in school settings with researchers as primary interventionists. Furthermore, these interventions were primarily implemented in non-inclusive environments, such as the resource room or an empty room within the school. A key limitation of this review was that the intervention outcomes were evaluated using only the rules of visual analysis. While the authors safeguarded their analyses by collecting 100% reliability during descriptive coding, visual analysis serves to determine functional relationships rather than causal relationships. Visual analysis does not allow for the identification of differential effects based on variations in study characteristics, such as participant characteristics and implementation differences. Assessing the efficacy of the intervention and the differential effects of the variables would require a meta-analysis.

Meta-analysis allows findings from multiple studies to be aggregated and statistically analyzed to identify potential moderators that produce differential effects (Richman et al., 2015). Several meta-analyses have examined the effects of challenging behavior interventions on individuals with DD (e.g., Bruni et al., 2017; de Bruin et al., 2013; Didden et al., 2006; Walker et al., 2018). However, these meta-analyses are limited in scope.

Didden et al. (2006) conducted a meta-analysis of research on behavioral interventions for individuals with mild intellectual disabilities. Eighty studies that included 109 participants (M age = 14.2 years; range 2 to 42 years old) met the criteria for the analyses in their study. The authors calculated two effect size (ES) measures: percentage of nonoverlapping data (PND, Scruggs et al., 1987) and percentage of zero data (PZD, Scotti et al., 1991). Results of their study indicated consequence-based, such as differential reinforcement strategies and extinction, and antecedent-based interventions, such as antecedent control, were the most effective for individuals with ID. These interventions were associated with a larger effect size than self-management procedures. Additionally, interventions based on experimental functional analysis yielded significantly stronger effects than interventions based on descriptive functional assessments. Although this meta-analysis discussed the effects of various types of behavioral interventions in reducing challenging behavior, further analysis is needed to determine whether these interventions are effective in

school settings and for adolescents with DD. Furthermore, this study did not evaluate the methodological rigor of the included studies prior to evaluating the effects of the interventions, which limits the believability of the findings.

More recently, Bruni et al. (2017) analyzed the effects of behavioral interventions used in school settings to reduce challenging behavior in students with and without disabilities. Moderator effects (i.e., presence and type of functional behavior assessment (FBA), classification, classroom placement, intervention setting) were measured through the calculation of Tau-U indices (Parker et al., 2011). Thirty studies with a total of 83 participants were included for analysis. Only 28 participants were adolescents, and it was not clear how many of the adolescents were typically developing or identified with DD. While this meta-analysis showed that behavioral intervention strategies had a moderate effect in reducing challenging behavior for both students with and without disabilities, this meta-analysis did not examine research quality and disaggregated according to the type of disability or age. Further analyses are required to determine whether severity, DD diagnosis, and types of interventions affect the efficacy of behavioral interventions for adolescents with DD. No other variables were found to moderate the effects of interventions.

Walker et al. (2018) examined the effects of function-based interventions in reducing challenging behavior of students with disabilities in inclusive school settings using nonoverlap of all pairs (NAP; Parker & Vannest, 2009) and Tau-U. Only studies with sufficient methodological rigor as evaluated through the WWC standards for single-case research (Kratochwill et al., 2010; WWC, 2020) were included in the analyses. Overall, results indicated that function-based interventions had moderate to strong effects in reducing challenging behavior of students with disabilities. Consistent with findings from prior meta-analyses (Bruni et al., 2017; Walker & Snell, 2013), the authors found that the type of functional behavioral assessment did not yield significant differences in behavioral outcomes. However, the reduction in challenging behavior was more significant when teachers implemented assessments and interventions compared to researchers and therapists. These findings are promising given that educators have the responsibility to provide academic instruction and effectively address challenging behavior in the classroom. Although this meta-analysis did not have age restrictions for participants, the studies that met criteria for analysis only included students from early childhood to elementary school, and the age of the students was not clearly specified. Furthermore, this meta-analysis did not disaggregate the results according to age, type of developmental disability (i.e., comparisons were made between students with DD and students with learning disability), type of behavioral intervention, or compare the effects of function-based and nonfunction-based interventions on the reduction of challenging behavior.

Only one study has quantified the effects of school-based behavioral interventions for adolescents with autism. de Bruin et al. (2013) examined the overall effects of four intervention categories (i.e., antecedent-, consequence-, self-management-, and video-based practices) on aggregated student outcomes, including challenging behavior, for adolescents with ASD (aged 12 – 22 years old). Thirty-four studies with a total of 99 adolescent participants met criteria for analysis in this study. Effect sizes (i.e., Phi, ϕ) were derived using the percentage of all nonoverlapping data (PAND, Parker et al., 2007). Results of their study reported moderate to large

weighted effects for antecedent- ($\phi = 0.67$), consequence- ($\phi = 0.75$), and video-based ($\phi = 0.90$) interventions. The effect sizes calculated for self-management interventions were generally strong ($\phi = 0.60 - 0.98$). However, there were not enough studies to provide a meaningful conclusion regarding the effects of self-management strategies. While this meta-analysis examined school-based behavioral interventions for adolescents with ASD, moderator analyses and examination of research quality were not conducted.

Although the results of previous meta-analyses provide initial support for the efficacy of behavioral interventions in the treatment of challenging behavior among adolescents with DD, several gaps in the literature remain. First, previous meta-analyses on challenging behavior interventions have been population-specific (de Bruin et al., 2013), did not include adolescents, or did not disaggregate data by participant age (Bruni et al., 2017; Didden et al., 2006; Walker et al., 2018), making it impossible to isolate intervention effects for adolescents with DD. Second, most of the meta-analyses on this topic did not review the methodological quality of studies, limiting the trustworthiness of the statistical analyses. Third, previous meta-analyses have examined the effects of specific interventions but did not determine if other variables (e.g., intervention type, participant and study characteristics, intervention components) moderate intervention efficacy. Last, since the publication of many of the previous meta-analyses, new effect size measures have been developed for single-case experimental research.

While the field of behavioral sciences currently lacks a standardized method for calculating ES, Tau-U has become one of the most popular and precise effect size measures for single-case research. Tau-U is a nonparametric ES measure that calculates the percentage of nonoverlapping data points between adjacent phases. Tau-U has been shown to provide the greatest power and precision in comparison to other nonoverlapping methods of calculation (Parker et al., 2011). Tau-U controls for autocorrelation accounts for baseline trend, and is robust against ceiling effects, unlike other effect size measures (Parker et al., 2011). Further, Tau-U has been used as the effect size metric in several meta-analyses in school-based interventions (e.g., Bruni et al., 2017; Walker et al., 2018).

While previous meta-analyses provide some evidence for the efficacy of challenging behavior interventions for adolescents with DD, several major gaps in the literature warrant additional investigation. Therefore, the purpose of the current meta-analysis was to extend previous research on challenging behavior interventions and to determine the efficacy of school-based challenging behavior interventions for adolescents with DD using the Tau-U ES measure. The following research questions were addressed:

(a) What is the overall effect of school-based behavioral interventions in reducing challenging behavior of adolescents with DD?
(b) Which interventions (i.e., antecedent-based, consequence-based, or combined) are considered effective in reducing challenging behavior of adolescents with developmental disabilities in school settings?
(c) Do participant (i.e., disability, verbal ability, type of challenging behavior) and study characteristics (i.e., classroom placement, implementation fidelity, inter-

ventionist, functional behavior assessment) yield differential effects on reduction in challenging behavior?

(d) Do intervention components (i.e., extinction, positive reinforcement, visual stimuli) affect the effectiveness of behavioral interventions in reducing challenging behavior of adolescents with developmental disabilities in school settings?

Method

This meta-analysis serves as an extension of the review by David et al. (2021). The method applied for the identification and selection of studies in the previous review is summarized below, including the literature search, inclusion criteria, and quality evaluation.

Literature Search

Articles included in this meta-analysis were based on the search procedures outlined by David et al. (2021). Four electronic databases were systematically searched to identify behavioral interventions targeting the reduction of challenging behavior and utilized in school settings for adolescents with developmental disabilities. These databases included Educational Resources Information Center (ERIC), Education Source, Education Full-Text, and PsycInfo. Variations of keywords from six categories (developmental disability, challenging behavior, school, intervention, reinforcement interventions, and antecedent interventions) were combined and entered into the electronic databases (see Table 1 on the review by David et al., 2021). The initial search was conducted in October 2017. Searches were updated in October 2018 and September 2020. A total of 6141 articles were screened for inclusion.

Study Selection

This meta-analysis utilized similar inclusion criteria as the previous review (David et al., 2021), however, to facilitate the calculation of effects, baseline data were also required for inclusion in the meta-analysis. Articles were screened for inclusion in this meta-analysis using the following seven criteria: (a) studies were published in English, peer-reviewed journals between 1997 and 2018 (the range was chosen to reflect the year in which IDEA was first reauthorized and promoted increased accountability among school personnel; David et al., 2021), (b) studies utilized a single-case experimental design (i.e., reversal/ABAB, multiple-baseline, alternating treatment; studies with changing criterion were not included for analysis) to demonstrate experimental control, (c) studies included a baseline for comparison of effects, (d) the study included at least one adolescent participant (i.e., aged 12 through 18 years old); participants described to be in sixth through 12th grade and/or as "adolescent" were also included, (e) the adolescent participants were reported as having DD, including autism and intellectual disability, such as Down syndrome, fragile X syndrome, (f) studies were implemented within a school setting (i.e., school was defined as any facility referred to as a "school"; residential and private programs (e.g., university clinics, day programs)

were not included in this meta-analysis, (f) the primary intervention consisted of a behavioral intervention targeting the reduction of challenging behavior, and (g) the dependent variable was a measure of challenging behavior. Studies that did not meet the described criteria were excluded from this meta-analysis. With the added inclusion criteria (i.e., baseline data requirement), 9 studies were removed from the 53 studies that were initially included in the previous review (David et al., 2021). Six additional articles met the inclusion criteria when the search was updated in September 2020, resulting in a total of 50 articles meeting the inclusion criteria.

Quality Evaluation

Articles that met criteria were further evaluated for their quality and rigor of methodology using an adapted standards of the Council for Exceptional Children (CEC) standards for evidence-based practices in special education (David et al., 2021). The CEC standards evaluate evidence quality using eight quality indicators: 1.0 context and setting, 2.0 participants, 3.0 intervention agent, 4.0 description of practice, 5.0 implementation fidelity, 6.0 internal validity, 7.0 outcome measures and 8.0 data analysis. Criteria for meeting these quality indicators were based on the descriptions provided by Cook et al. (2015). Operational definitions for each QI and related subindicators are described in the previous review (David et al., 2021). Each indicator consists of one to nine sub-indicators that must be met for the overall domain to be met. Only sub-indicators related to single-case research were utilized. Previous studies have shown that many studies do not meet the CEC criteria to be considered methodologically sound due to missing contextual information (i.e., QI 1.0 – 4.0; Andzik et al., 2016; Gregori et al., 2018) and implementation fidelity information (QI 5.0). The lack of some contextual information limits the generalizability of a given study (Andzik et al., 2016; David et al., 2021), while the lack of implementation fidelity information limits the confidence placed on any established relationships (David et al., 2021). However, missing contextual information does not impact how studies establish a functional relationship between independent (Andzik et al., 2016) and dependent variables, whereas the lack of reporting of implemenation fidelity does not automatically indicate that studies implemented studies with low fidelity or poor experimental control (David et al., 2021). Functional relationships and internal validity are examined using indicators 6.0 through 8.0. Therefore, studies that met indicators 6.0 through 8.0 were considered high-quality and analyzed for this meta-analysis. Thirty studies met quality indicators 6.0 – 8.0 and were included for analyses.

Variables Coded

Following the identification of studies for analysis, descriptive information was extracted from each study using the following variable categories: participant and setting characteristics (i.e., gender, developmental disability, verbal ability, classroom setting), study components (i.e., challenging behavior, functional behavior assessment and implementation components (i.e., interventionist, treatment fidelity, function-based intervention, intervention type and intervention components).

Participant and Setting Characteristics

Participant characteristics included gender, primary disability, and verbal ability. Gender was coded as either male or female. Primary disability was coded as either ASD, ID, or other DD. For individuals with multiple disabilities reported, their primary disability was identified as the first disability reported based on author-reported descriptions of the individuals. Verbal ability was coded as verbal (i.e., participants communicated primarily through verbal responses), limited verbal (i.e., participants communicated using limited verbal responses, such as one- to three-word responses), or nonverbal (i.e., participants communicated using gestures, vocalizations, or a speech-generating device). The setting variable was coded as either classroom environment (i.e., general education, self-contained, resource classrooms) or non-classroom environments (i.e., any other setting within the school facility, such as empty rooms, hallways, playgrounds, motor room).

Challenging Behavior Challenging behavior was categorically coded as aggressive or passive. Aggressive behaviors were defined as any behaviors that pose a threat to the individual or others (i.e., physical aggression toward self or others, self-injurious behaviors). Passive behaviors were defined as any behaviors that prevent the individual from accessing the academic curriculum or engaging appropriately with peers (e.g., motor or vocal stereotypy, disruptive vocalizations, off-task behaviors, and noncompliance). For individuals with multiple behaviors, we applied the following rules: (1) individuals were coded as having aggressive behavior if studies described the individuals to be engaging in at least one form of aggressive behavior, and (2) individuals were coded as having passive behavior if studies described the individuals to be engaging in only passive behaviors.

Study Components

Five variables were included in this category: functional behavior assessment, function-based intervention, interventionist, treatment fidelity, and intervention type.

Functional Behavior Assessment This variable was categorically coded as descriptive, experimental, or none. Descriptive assessments were defined as the assessment of behavior function through either direct observation of the behavior in the natural environment or indirect assessments, such as interviews, questionnaires, or rating scales designed to identify possible functions of behaviors (Gresham et al., 2001). Experimental assessment refers to the systematic manipulation of antecedents and consequences surrounding the target behavior (i.e., any variations experimental functional analysis; Gresham et al., 2001; Iwata et al., 1982; Scott & Kamps, 2007).

Function-Based Intervention Function-based intervention was coded as a binary system of "1" (i.e., the authors reported that the intervention was based on the results of an FBA) or "0" (i.e., the authors did not report that the intervention was based on the results of an FBA).

Interventionist Interventionist was coded as researcher, teaching personnel, or other personnel. Teaching personnel was coded if the interventionist was the teacher, paraprofessional, or any other member of the school facility. Other personnel were coded for individuals who were not clearly defined as being part of the research team or the school facility.

Treatment Fidelity Treatment fidelity was coded as a binary system of "1" (i.e., treatment fidelity was reported and assessed) or "0" (i.e., treatment fidelity was not reported and assessed).

Intervention Type Intervention type was coded as either antecedent-based, consequence-based, or combined. Antecedent-based interventions were any interventions in which a component of the environment was altered to occasion the occurrence of appropriate behaviors and decrease the likelihood of challenging behavior through the manipulation of motivating operations (e.g., presession satiation, providing choices, environmental arrangements, high-probability response sequence, or high-p), increasing predictability of future events (i.e., visual schedules) and teaching classroom expectations (i.e., social stories). Consequence-based interventions were any interventions that relied on (a) increasing availability of reinforcers either noncontingently or contingent upon engaging in appropriate behaviors (i.e., variations of differential reinforcement, including functional communication training) or (b) alteration of stimuli to decrease the likelihood of engaging in challenging behavior using punishment procedures (e.g., response blocking, physical restraint). Combined interventions were any interventions that included both antecedent- and consequence-based interventions.

Intervention Components

Several components were initially examined. However, only the following components had reasonable variation to allow for group comparisons: visual stimuli, replacement behaviors, extinction, and planned reinforcement. These variables were coded as a binary system of "1" (i.e., the intervention component was part of the intervention) or "0" (i.e., the intervention component was not part of the intervention). Visual stimuli were defined as the use of any form as visual cues or directions, such as activity schedules, written directions, or communication boards, as an intervention component to reduce the challenging behavior of the target. Extinction was defined as systematic termination of reinforcement delivery following occurrence of target challenging behavior. Replacement behaviors were defined as teaching the target participant an alternative behavior to replace the target challenging behavior as an intervention component. Planned reinforcement was defined as any schedule of reinforcement systematically planned as part of the intervention (e.g., during FCT, reinforcement is delivered every time an individual produces the target mand).

Data Extraction

Intervention data relevant to study purposes were extracted from each study using WebPlotDigitizer, an empirically validated and reliable computer program (Rohatgi, 2021; Drevon et al., 2017). Graphs were extracted from all included studies. For each study, the screenshot of the graph is uploaded into the program. The X and Y axes are calibrated accordingly. Each data point is manually selected to generate the corresponding numerical X and Y values. Data extracted from WebPlotDigitizer were imported into a Microsoft Excel file and organized by phase (i.e., baseline or intervention) for further analysis. See Drevon et al. (2017) for a detailed task analysis on how to extract data using WebPlotDigitizer.

Interrater Agreement

Inter-rater agreements (IRA) were calculated by calculating the overall percent agreement. IRA were measured by dividing the number of disagreements by the total number of agreements plus disagreements and multiplying by 100. Doctoral students in a special education program and undergraduate students were trained by the first author as second reviewers using a researcher-developed checklist and protocol for the following procedures: (a) literature search, (b) quality evaluation, (c) data extraction, and (d) moderator coding. Each rater was required to meet the training criterion of 80% or higher agreement for two consecutive articles. In cases of disagreements, the rater met and discussed the disagreements until 100% agreement was obtained.

Literature Search

Undergraduate and doctoral students independently evaluated 30% ($n = 809$) of the resulting articles as second reviewers against the inclusion criteria. The IRA for the inclusion and exclusion of articles was 96% (range 92% – 100%).

Quality Evaluation Doctoral students were trained using a researcher-developed protocol as reviewers to independently assess 94% of the included articles. The IRA for the quality evaluation averaged 94% (range 90%—100%).

Data Extraction A graduate student was trained to extract intervention data using WebPlotDigitizer for 34% of the included articles. Agreement was defined as having a difference of no more than 1 between the extracted X and Y values. The IRA for data extraction was 98% (range 75% – 100%). The large range of agreement is due to a calibration error between the coders for one of the studies. All the x-axis data points were aligned; however, several y-axis points were beyond the margin of error, predetermined by the first author.

Moderator Coding Undergraduate and graduate students independently reviewed the accuracy of the extracted descriptive variables for 100% of the included articles.

Agreement was defined as each rater extracting the same variable information. The IRA for moderator coding was 98% (range 88%—100%).

Data Analyses

Phase Contrast Selection

Only adjacent baseline and intervention phases (i.e., A-B contrast) were considered for analyses. Each baseline and intervention combination counted as one demonstration of effect and was evaluated with an ES. To ensure the independence of phase contrasts, each baseline and intervention were evaluated only once. Therefore, only phases where intervention immediately followed baseline were evaluated and any subsequent intervention phases were not included in the analyses.

Effect Size

Tau-U was used as the effect size measure from the extracted studies (Parker et al., 2011). Tau-U is a nonparametric measure of effect size that is measured by comparing nonoverlapping data between two phases (i.e., all data points from Phase A are compared to all data points in Phase B; Parker et al., 2011). Reverse rank-ordered data (e.g., the highest data point would be ranked as one and as the value of the data decreases, the rank order increases) were used to calculate the effect size without correcting the baseline trend. Data were extracted from each A-B contrast (e.g., baseline vs. intervention). Based on Parker and Vannest's (2009) guidelines for interpreting Tau-U, the effect size is classified as small if the effect size ranged between 0 and 0.65, moderate if the effect size ranged between 0.66 to 0.92, and large if the effect size ranged between 0.93 and 1.00.

Effect Size Aggregation

Omnibus effect sizes were also determined by aggregating the ES across all included studies in addition to calculating the differences across moderators. Similar methods used in aggregating cases were used to obtain an aggregated Tau-U. A mean Tau-U is obtained after each ES is weighted by the inverse of the variance score (Var_s).

Comparison of Effects

Participant and intervention components were coded categorically to examine differing levels of effects. A Wilcoxon signed-rank statistic (Wilcoxon, 1945) was used to determine statistical significance for variables with only two categories. A Kruskal–Wallis one-way analysis of variance (Kruskal & Wallis, 1952) was used to determine statistical significance for variables with three of more categories. Further, if the results of the Kruskal–Wallis test showed statistical significance, a Dunn post-hoc test (Dunn, 1964) was used to examine the significance across each pairwise group combination.

Results

Quality Evaluation

The 50 studies that met inclusion criteria were quality evaluated using the CEC standards (Cook et al., 2015). Of the 50 studies that met inclusion criteria, 30 studies met criteria for analysis (i.e., QIs 6.0 – 8.0 were met). Interestingly, only three studies met all CEC QIs (1.0 – 8.0). Six studies met QIs related to reporting of specific contextual information (QIs 1.0 – 4.0) and of those six studies, only four studies reported information regarding implementation fidelity (QI 5.0). Across the 50 studies, the quality indicators to which most of the studies met were QI 1.0 context and setting (96%), QI 4.0 description of practice (100%), QI 7.0 outcome measures (88%), and QI 8.0 (100%). The QIs that studies met the least were QI 2.0 participants (30%), QI 3.0 intervention agent (42%), QI 5.0 implementation fidelity, and QI 6.0 internal validity. Further description of the quality evaluation results were described in the review by David et al. (2021).

Descriptive Synthesis

Sixty-one ES were extracted in 30 studies meeting CEC QIs 6.0 through 8.0 with 45 adolescent participants. The omnibus Tau-U ES for behavioral interventions targeting the reduction of challenging behavior was 0.68 ($SD = 0.04$, [0.62, 0.73]). These studies were also descriptively reviewed using moderators associated with the participant, setting, challenging behavior, functional behavior assessment, and intervention components. Descriptive summary of the studies that met criteria for analyses is reflected in Table 1. Forest plot of group Tau-U ES and confidence intervals (CI) across all potential moderators are reflected in Fig. 1.

Participant and Setting Characteristics

Effect sizes were calculated across the following participant and setting variables: primary disability, comorbidity, verbal ability, and setting.

Gender and Primary Developmental Disability Forty-five participants were included for analyses. Approximately 67% of the participants ($n = 30$) were male. Of the 45 participants, 26 participants (58%) have a primary diagnosis of ASD, 19 participants (42%) have ID and 1 participant was categorized as having other DD (i.e., Down syndrome). Aggregate Tau-U was 0.68 ($SD = 0.05$, CI_{95} [0.61, 0.74]) for participants with ASD and 0.67 ($SD = 0.06$, CI_{95} [0.58, 0.76]) for participants with ID; the number of participants with other DD ($n = 1$) was insufficient for obtaining an aggregate Tau-U. The test for statistical significance showed no significant differences between groups based on primary DD (Kruskal–Wallis $p = 0.66$).

Table 1 Descriptive Summary of Included Studies across Moderators

Study	Participant Characteristics					Implementation & Intervention Components						
	N of Subjects	Primary DD	Comorbidity	Verbal Ability	Challenging Behavior	FBA (FBI)	Interventionist	Intervention Type	Visual Stimuli	Replacement Behavior	Extinction	Planned Reinforcement
Banda and Kubina (2006)	1	ASD	Single	V	Passive	None (No)	Typical Agent	Antecedent	N	N	N	N
Banda and Kubina (2009)	1	ASD	Single	-	Passive	None (No)	Researcher	Antecedent	N	N	N	N
Banda et al. (2012)	1	ASD	Multiple	NV	Aggressive	Experimental (Yes)	Typical Agent	Combined	N	N	Y	Y
Bilial-Lolis et al. (2012)	3	ID; ASD; ID	Single; Multiple; Single	LV	Aggressive	None (No)	Researcher	Antecedent	Y	N	N	Y
Brown and Mirenda (2006)	1	ASD	Single	LV	Passive	Experimental (Yes)	Typical Agent	Combined	Y	Y	N	Y
Butler et al. (2007)	1	ASD	Single	LV	Aggressive	Experimental (Yes)	Researcher	Combined	N	N	Y	Y
Chung et al. (2010)	1	ID	Single	NV	Aggressive	Experimental (Yes)	Researcher	Antecedent	N	N	N	Y
Clarke et al. (2002)	1	ASD	Multiple	LV	Aggressive	Descriptive (Yes)	Typical Agent	Combined	Y	N	N	Y
Courtemanche et al. (2014)	1	ASD	Single	NV	Aggressive	Descriptive (Yes)	Typical Agent	Consequence	N	Y	Y	Y
Dibley et al. (1999)	1	ID	Single	LV	Passive	Descriptive (Yes)	-	Antecedent	N	N	N	N
Dozier et al. (2007)	1	ID	Single	LV	Aggressive	Experimental (Yes)	-	Combined	Y	Y	N	Y

Table 1 (continued)

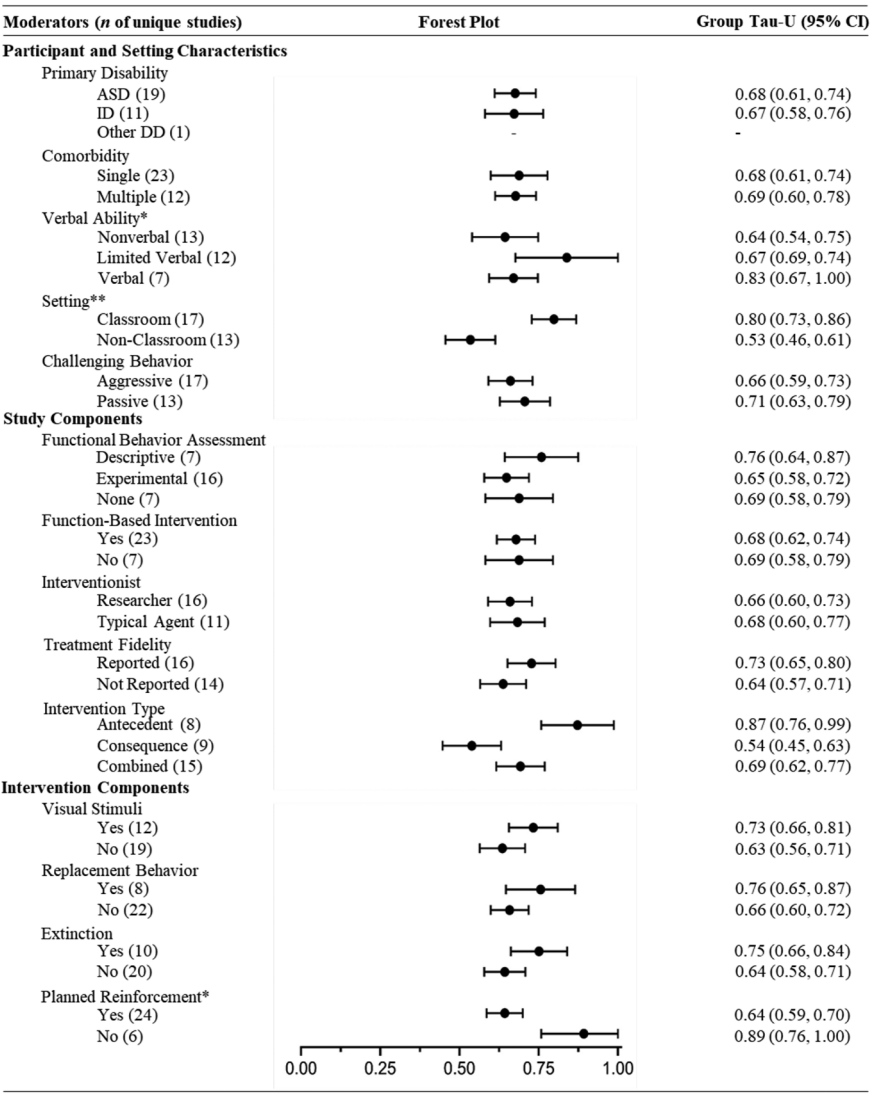
Study	Participant Characteristics					Implementation & Intervention Components						
	N of Subjects	Primary DD	Comorbidity	Verbal Ability	Challenging Behavior	FBA (FBI)	Interventionist	Intervention Type	Visual Stimuli	Replacement Behavior	Extinction	Planned Reinforcement
Durand (1999)	1	ID	Multiple	NV	Passive	Experimental (Yes)	Typical Agent	Combined	Y	Y	N	Y
Gibney et al. (2020)	4	ASD	Single	V; LV; NV	Passive	Experimental (Yes)	Researcher	Antecedent; Consequence	N	N	N	Y
Hammond et al. (2011)	2	ASD	Multiple; Single	-	Aggressive	Experimental (Yes)	Researcher	Consequence; Combined	Y	N	Y	Y
Hirasawa et al. (2009)	2	ASD	Single	NV	Aggressive	Descriptive (Yes)	-	Combined	N	N	Y	Y
Kim et al. (2014)	3	ID	Single	LV	Passive	Descriptive (Yes)	Researcher	Combined	Y	N	N	Y
Machalicek et al. (2009)	1	ASD	Single	LV	Passive	None (No)	Typical Agent	Combined	Y	N	N	Y
Muething et al. (2018)	2	ASD	Multiple	V; NV	Aggressive	Experimental (Yes)	Researcher	Combined	Y	Y	Y	Y
O'Reilly et al. (2005)	1	ASD	Multiple	NV	Aggressive	Experimental (Yes)	Researcher	Combined	N	N	N	Y
Pipkin et al. (2010)	1	ID	Single	NV	Aggressive	Experimental (Yes)	Researcher	Consequence	N	Y	Y	Y
Rogalski et al. (2020)	1	ASD	Single	LV	Aggressive	Experimental (Yes)	Researcher	Consequence	N	N	N	Y
Rosenbloom et al. (2019)	1	ASD	Single	-	Passive	None (No)	Researcher	Antecedent	Y	N	N	N
Scattone et al. (2002)	1	ASD	Single	V	Passive	None (No)	Typical Agent	Antecedent	Y	N	N	N
Schmidt and Stichter (2012)	1	ID	Multiple	NV	Passive	Descriptive (Yes)	Typical Agent	Consequence	N	N	N	Y

Table 1 (continued)

Study	Participant Characteristics					Implementation & Intervention Components						
	N of Subjects	Primary DD	Comorbidity	Verbal Ability	Challenging Behavior	FBA (FBI)	Interventionist	Intervention Type	Visual Stimuli	Replacement Behavior	Extinction	Planned Reinforcement
Slaton et al. (2017)	3	ASD	Single; Multiple; Single	V; LV	Aggressive	Experimental (Yes)	Researcher	Consequence	N	Y	Y	Y
Stuesser and Roscoe (2020)	2	ASD	Multiple; Single	NV; V	Aggressive	Experimental (Yes)	Researcher	Consequence	N	N	N	Y
Tomlin et al. (2012)	1	Other DD	Multiple	V	Passive	Descriptive (Yes)	Typical Agent	Consequence	N	N	Y	Y
VanCamp et al. (2000)	1	ID	Single	NV	Aggressive	Experimental (Yes)	Researcher	Consequence	N	N	N	Y
Vollmer et al. (1999)	1	ID	Single	NV	Aggressive	Experimental (Yes)	Researcher	Consequence	N	Y	Y	Y
Whatley et al. (2009)	4	ID	Multiple; Multiple; Single; Single	LV	Passive	None (No)	Typical Agent	Combined	Y	N	N	Y

ASD autism spectrum disorder, *ID* intellectual disability, *Other DD* other developmental disability, *V* verbal, *LV* limited verbal, *NV* nonverbal, *FBA* functional behavior assessment, *FBI* function-based intervention, *Y* yes, *N* no

Forest Plot of Group Effect Sizes Across Moderators



*Note: **p < .01 level; *p < .05 level; ASD =autism spectrum disorder; DD = developmental disability; ID = intellectual disability.*

Fig. 1 Forest Plot of Group Effect Sizes Across Moderators

Comorbidity Of the 45 participants, 31 participants (69%) reported having only one DD while 14 participants (31%) were reported to have a comorbid disability. Aggregate Tau-U were 0.68 ($SD = 0.05$, CI_{95} [0.61, 0.74]) for participants with a single disability and 0.69 ($SD = 0.06$, CI_{95} [0.60, 0.78]) for participants with a reported comorbid disability. No statistically significant differences were found across groups based on comorbidity (Wilcoxon $p = 0.49$).

Verbal Ability Information on verbal ability was extracted for 43 participants (96%). Verbal ability information was not available for two participants (4%). Of the 43 participants, 13 participants (29%) were nonverbal or primarily communicated using gestures, 21 participants (47%) had limited verbal ability, and 7 participants (16%) primarily communicated verbally. Aggregate Tau-U were 0.64 ($SD = 0.07$, CI_{95} [0.54, 0.75]) for participants who were nonverbal, 0.67 ($SD = 0.05$, CI_{95} [0.59, 0.74]) for participants with limited verbal ability, and 0.83 ($SD = 0.11$, CI_{95} [0.67, 1]) for participants who primarily verbally communicated. The Kruskal–Wallis test of significance ($p = 0.02$) indicated statistically significant differences between the groups based on verbal ability, with stronger effects shown for participants who were verbal.

Setting Of the 30 studies, intervention was implemented within a classroom environment for 18 studies and a non-classroom environment for 12 studies. The aggregate Tau-U was 0.80 ($SD = 0.05$, CI_{95} [0.73, 0.86]) for participants receiving intervention within classroom environment and the aggregate Tau-U was 0.53 ($SD = 0.05$, CI_{95} [0.63, 0.79]) for participants receiving intervention in a non-classroom environment. The test for statistical significance showed significant differences between groups based on the intervention setting (Wilcoxon $p < 0.01$).

Challenging Behavior Among the 45 participants, 24 participants (53%) engaged in aggressive challenging behavior, and 21 participants (47%) engaged in passive challenging behavior. Aggregate Tau-U were 0.66 ($SD = 0.05$, CI95 [0.59, 0.73]) for participants engaging in aggressive challenging behavior and 0.71 ($SD = 0.06$, CI_{95} [0.63, 0.79]) for participants engaging in passive challenging behavior. The test for statistical significance showed no significant differences between groups based on the type of challenging behavior (Wilcoxon $p = 0.29$).

Study Components

Effect sizes were calculated for the following implementation variables: functional behavior assessment, function-based interventions, interventionist, treatment fidelity, and intervention type.

Functional Behavior Assessment Sixteen out of the 30 studies conducted an experimental analysis of challenging behavior, whereas seven studies conducted only a descriptive analysis of challenging behavior. Seven studies did not report conducting any type of FBA. Aggregate Tau-U were 0.65 ($SD = 0.05$, CI_{95} [0.58, 0.72]) for studies that conducted an experimental analysis of challenging behavior, 0.76 ($SD = 0.08$, CI_{95} [0.64, 0.87]) for studies that conducted a descriptive analysis of challenging behavior, and 0.69 ($SD = 0.07$, CI_{95} [0.58, 0.79]) for studies that did not conduct any FBA. No statistically significant differences were found across groups based on FBA (Kruskal–Wallis $p = 0.96$).

Function-Based Intervention Twenty-three studies reported developing the intervention based on the outcomes of the FBA while seven studies did not report conducting any FBA prior to developing the intervention. Aggregate Tau-U were 0.68 ($SD = 0.04$, CI_{95} [0.62, 0.74]) for studies implementing function-based interventions and 0.69 ($SD = 0.07$, CI_{95} [0.58, 0.79]) for studies implementing non-function-based interventions. No statistically significant differences were found across groups based on the type of intervention (Wilcoxon $p = 0.40$).

Interventionist Of the 30 studies, 16 studies utilized researchers (53%) and 11 studies utilized typical agents (37%) as interventionists. The interventionists could not be clearly determined in three studies (10%). Aggregate Tau-U were 0.66 ($SD = 0.05$, CI_{95} [0.60, 0.73]) for studies utilizing researchers and 0.68 ($SD = 0.06$, CI_{95} [0.60, 0.77]) for studies utilizing typical agents as interventionists. No statistically significant differences were found across groups based on interventionist (Wilcoxon $p = 0.20$).

Treatment Fidelity Sixteen out of the 30 studies reported treatment fidelity (53%) whereas 14 studies did not report treatment fidelity (47%). Aggregate Tau-U were 0.73 ($SD = 0.05$, CI_{95} [0.65, 0.80]) for studies reporting treatment fidelity and 0.64 ($SD = 0.05$, CI_{95} [0.57, 0.71]) for studies that did not report treatment fidelity. No statistically significant differences were found across groups based on the reporting of treatment fidelity (Wilcoxon $p = 0.20$).

Intervention Type Across the 30 studies, eight studies utilized antecedent-based interventions, nine studies utilized consequence-based interventions, and 15 studies utilized interventions utilizing both antecedent- and consequence-based interventions. Aggregate Tau-U were 0.87 ($SD = 0.08$, CI_{95} [0.76, 0.99]) for studies implementing antecedent-based interventions, 0.54 ($SD = 0.07$, CI_{95} [0.45, 0.63]) for studies implementing consequence-based interventions, and 0.69 ($SD = 0.05$, CI_{95} [0.62, 0.77]) for studies implementing a combination of antecedent- and consequence-based interventions. No statistically significant differences were found across groups based on the type of intervention (Kruskal–Wallis $p = 0.05$).

Intervention Components

A binary system (i.e., 1 = included or 0 = not included) was utilized to determine the presence or absence of the following intervention components: visual stimuli, replacement behavior, extinction, and planned reinforcement.

Visual Stimuli Twelve studies (40%) included visual stimuli as an intervention component, whereas 19 studies (63%) did not include visual stimuli as an intervention component. Aggregate Tau-U were 0.73 ($SD = 0.05$, CI_{95} [0.66, 0.81]) for studies that included visual stimuli as an intervention component and 0.63 ($SD = 0.04$, CI_{95} [0.56, 0.71]) for studies that did not include visual stimuli as an intervention

component. No statistically significant differences were found across groups based on utilizing visual stimuli as an intervention component (Wilcoxon $p = 0.47$).

Replacement Behavior Eight studies (27%) included replacement behavior as an intervention component, whereas 18 studies (73%) did not include replacement behavior as an intervention component. Aggregate Tau-U were 0.76 ($SD = 0.08$, CI_{95} [0.65, 0.87]) for studies that included replacement behavior as an intervention component and 0.66 ($SD = 0.04$, CI_{95} [0.60, 0.72]) for studies that did not include replacement behavior as an intervention component. No statistically significant differences were found across groups based on utilizing replacement behavior as an intervention component (Wilcoxon $p = 0.20$).

Extinction Ten studies (33%) included extinction as an intervention component, whereas 20 studies (67%) did not include extinction as an intervention component. Aggregate Tau-U were 0.75 ($SD = 0.06$, CI_{95} [0.66, 0.84]) for studies that included extinction as an intervention component and 0.64 ($SD = 0.05$, CI_{95} [0.58, 0.71]) for studies that did not include extinction as an intervention component. No statistically significant differences were found across groups based on utilizing extinction as an intervention component (Wilcoxon $p = 0.12$).

Planned Reinforcement Twenty-four studies (80%) included planned reinforcement as an intervention component, whereas six studies (20%) did not include planned reinforcement as an intervention component. Aggregate Tau-U were 0.64 ($SD = 0.05$, CI_{95} [0.59, 0.70]) for studies that included planned reinforcement as an intervention component and 0.89 ($SD = 0.09$, CI_{95} [0.76, 1.00]) for studies that did not include planned reinforcement as an intervention component. The test for statistical significance showed significant differences across groups based on utilizing planned reinforcement as intervention component (Wilcoxon $p < 0.01$).

Discussion

The purposes of this meta-analysis were to examine the effects of school-based interventions on reducing challenging behavior of adolescents with developmental disabilities and to determine whether specific participant, study, and intervention characteristics moderated such effects. Thirty studies that met the CEC quality indicators focusing on methodological rigor were included for analysis. The use of studies that showed evidence of rigorous methodology increased confidence in the results, though these results should be interpreted with caution given the limited number of included studies. Overall, findings from this meta-analysis fill several gaps in the literature on school-based behavior interventions for adolescents with DD and provide several implications for practice and research.

Overall Effects of School-Based Interventions

A moderate effect size (Tau-U = 0.68, *SD* = 0.04, [0.62, 0.73]) was found in the analysis of the 30 included studies and across 61 unique contrasts, suggesting that behavioral interventions are generally effective in reducing challenging behavior of adolescents with DD in school-based settings. This finding is promising as adolescents with DD are generally underrepresented in school-based intervention research (Walker et al., 2018). Since challenging behavior among adolescents has been attributed to negative post-school outcomes (Wehman et al., 2015), it is pertinent that these behaviors are reduced prior to transitioning out of secondary education. However, the overall ES observed is comparatively lower to the overall effect size found in similar studies that primarily included younger children (e.g., Tau-U = 0.91, Walker et al., 2018). This finding suggests that there are likely additional factors (e.g., reinforcement history, severity of behavior) that affect the effectiveness of challenging behavior interventions among adolescents with DD.

Moderator Analyses of Effects

Moderate effects were observed in most moderators, suggesting that school-based challenging behavior interventions are generally effective in addressing varying degrees of challenging behavior in adolescents with DD. Several key findings resulted from the moderator analyses. First, majority of moderators examined did not yield statistically significant group differences, including intervention type. Second, only three moderators (i.e., verbal ability, setting, and planned reinforcement) were found to moderate intervention effects with statistical significance. Third, statistically significant differences were not observed between studies grouped based on pre-treatment functional assessment conducted prior to intervention delivery.

Participant and Setting Characteristics

Statistically higher reductions in challenging behavior were observed for individuals who primarily communicated verbally in comparison to individuals with limited verbal ability or individuals who were nonverbal. Verbal ability is potentially indicative of the severity of their disability, suggesting that behavior reduction strategies are likely to be more effective for those with a milder forms of DD. This is supported by prior research that indicates a strong relationship between disability severity and higher levels of challenging behavior (Levy & Perry, 2011; Ruiter & Dekker, 2008) and lower gains in behavioral domains for adolescents and adults with more severe forms of DD (Carter et al., 2012). However, as previously mentioned, this finding should be interpreted with caution as verbal ability alone is not a sufficient measure of disability severity. Furthermore, verbal ability was based on the author's report rather than specific clinical measures. This could lead to either a potential underestimation or an overestimation of actual verbal ability. Further analysis is required to closely determine how the severity of the disability, which should be based on

a measure of several domains, impacts outcomes of behavior reduction strategies. In contrast, no additional participant variables (i.e., primary disability, comorbidity, and challenging behavior) were observed to moderate the effects in intervention outcomes, aligning with previous research, suggesting that behavioral interventions can be effective in reducing challenging behavior regardless of disability or type of challenging behavior (Bruni et al., 2017; Walker et al., 2018).

In terms of intervention setting, statistically larger treatment effects were observed for interventions that were implemented within the students' typical classroom setting in comparison with other instructional settings within the schools (e.g., separate therapy rooms, gym), which is consistent with the findings by Bruni et al. (2017). This finding demonstrates that interventions are more likely to be effective when interventions are implemented within the relevant context.

Study Components

One of the major findings in this meta-analysis is that none of the variables relating to study components (i.e., FBA, function-based intervention, interventionist, treatment fidelity, intervention type) wasfound to moderate the effects of behavior interventions in reducing challenging behavior of adolescents with DD. Of this major finding, there were two noteworthy findings that warrant further discussion.

First, results of the moderator analysis did not indicate that the type of pre-treatment functional assessment (i.e., descriptive, experimental, or no FBA) moderated the effects of the intervention. This finding is consistent with previous meta-analyses (e.g., Bruni et al., 2017; Walker et al., 2018), suggesting that the effects of school-based behavioral interventions were similarly effective regardless of whether the intervention was based on descriptive FBA, experimental FBA, or reportedly not conducting an FBA. However, as the prior literature suggests, there are mixed findings regarding the impact of pretreatment functional assessments on treatment outcomes. Findings from this moderator analysis reflect a lack of differential effects across groups, which suggests that failing to conduct a pre-treatment functional assessment may not necessarily lead to poor student outcomes. However, results from this meta-analysis also suggest that conducting an FBA prior to treatment could lead to an effective intervention more efficiently as the target intervention matched the function of the challenging behavior. There are several factors that could account for this lack of differential effects. First, it is plausible that for some studies, an FBA was conducted but was simply not reported. Second, publication bias may be contributing to the finding, with published studies generally leaning towards positive findings. Third, as Bruni et al. (2017) noted, it is likely that FBA may not be necessary in addressing less intensive challenging behavior but would be recommended as the severity of challenging behavior intensifies. There is a potential linear relationship between the type of FBA and the severity of challenging behavior. As the severity of challenging behavior increases, the type of FBA recommended also increases in intensity. As such, using more intensive forms of FBA may be more efficient in determining effective interventions, especially for severe forms of challenging behavior that may lead to elevated risks of harm towards the individual and those who are in contact with the individual. However, this raises a particular concern for

practitioners regarding how to determine when to utilize a more intense FBA. Lastly, the lack of significant differences between groups may be influenced by the type of intervention. All studies ($n=7$) that did not conduct a pre-treatment functional assessment utilized antecedent-based strategies (Banda et al., 2006, 2009; Bilias-Lolis et al., 2012; Machalicek et al., 2009; Rosenbloom et al., 2019; Scattone et al., 2002; Whatley et al., 2009). Antecedent-based strategies serve as preventive measures and are known to positively impact challenging behavior (Radley & Dart, 2016; Smith, 2011). As a preventive measure, these strategies work by manipulating variables that occasion challenging behavior, potentially eliminating the need for knowledge of the consequence maintaining the challenging behavior.

Second, there were no statistically significant differences between groups based on intervention type (i.e., antecedent-based, consequence-based, and combined interventions). This finding is encouraging and suggests that educators have multiple intervention options to select from when designing challenging behavior interventions. Access to multiple intervention options also gives teachers freedom to select practices that align with their ongoing routines and fit within the existing classroom structure, which may increase teacher buy-in and long-term use of EBPs. As such, there are several interventions (e.g., FCT, differential reinforcement of alternative behavior) that are readily available and have been established as evidence-based practices for reducing challenging behavior.

Intervention Components

This meta-analysis examined whether including the following components, visual stimuli, replacement behavior, extinction, and planned reinforcement, contributed to the effects of the intervention. However, any of the findings discussed should be treated with caution. Several of the findings indicate that there were no differences between the effects of the interventions regardless of whether specific components were included or excluded from the intervention. These findings do not suggest that these components should be readily omitted as the purpose of this meta-analysis was not to compare whether interventions are more or less effective with the presence or absence of specific components. Findings of this meta-analysis indicate that there were no differences in intervention effects when visual stimuli, replacement behavior, and extinction were part of the intervention. However, when planned reinforcement was not included as an intervention component, statistically significant effects were observed.

Interestingly, statistically significant effects were found in studies that did not include planned reinforcement as an intervention component. Although group differences were found to be statistically significant, caution is needed in interpreting this finding due to the large discrepancy between the number of studies that did ($n=24$) and did not ($n=6$) include planned reinforcement as an intervention component. This finding could potentially be of false significance and warrant further exploration. However, there are few explanations that could explain this finding. First, the lack of significant effects for planned reinforcement could be due to differences in reinforcement contingencies during programmed arrangements and naturally occurring

arrangements. If the probability of the target behavior encountering reinforcement during naturally occurring arrangements is higher, the value of the reinforcer could decrease during programmed arrangements, especially if there is also an increase in response effort to obtain reinforcement. Second, of the six studies that did not report including planned reinforcement as an intervention component were four studies that implemented antecedent-based interventions (i.e., high-p (Banda & Kubina, 2006, 2009), choice making (Dibley & Lim, 1999), self-modeling (Bilias-Lolis et al., 2012), and social stories (Kim et al., 2014; Scattone et al., 2002). Antecedent-based interventions decrease the likelihood of challenging behavior occurrence by modifying environmental factors in such that these factors no longer evoke challenging behavior (Cooper et al., 2007). As such, these interventions may not necessarily require the addition of consequent-based procedures, such as planned reinforcement, suggesting that antecedent-based strategies may be sufficient in reducing challenging behavior.

However, as previously discussed, there was actually a great number of studies that did not include any information regarding implementation fidelity. Several studies did not report implementation fidelity, which could mean that intervention agents were providing reinforcers that were not part of the original study procedures. Additionally, not all of the studies that implemented planned reinforcement conducted preference or reinforcer assessments. This means that the stimuli selected as reinforcers may not have been functioning as reinforcers. Overall, this finding is preliminary, but potentially encouraging for practitioners because to some extent, antecedent-based procedures may be sufficient to reduce challenging behavior of students with DD. Antecedent-based interventions may be more cost-effective and less resource-intensive as resources may often be limited within school settings. Additional research is needed to determine the contingencies and components that contribute to the effectiveness of these antecedent-based strategies.

As with all studies, there are some limitations to be considered. First, this meta-analysis included studies that did ($n = 16$) and did not report implementation fidelity ($n = 14$). The lack of implementation fidelity data is a limitation for this meta-analysis as it limits the confidence that can be placed towards any conclusions made regarding the implementation of the intervention and treatment effects. Implementation fidelity is important to confidently conclude that the effects of the interventions are due to the independent variables rather than some other extraneous variable (McIntyre et al., 2007). Intervention must be conducted with full fidelity to confidently conclude that the effects of the interventions directly resulted from the implementation of the intervention and the intervention was implemented as intended. However, for this study, inclusion for analysis was primarily determined based on whether the study had strong experimental control. Based on currently available standards (e.g., CEC guidelines, WWC standards), there is a discrepancy on whether implementation fidelity should be considered as a determining factor of study quality. Furthermore, previous meta-analyses have reported that study quality did not impact intervention outcomes (Walker et al., 2018) or has not included study quality as a factor for inclusion and analysis (Bruni et al., 2017).

Second, the interventions included in this meta-analysis were grouped into collective groups (antecedent-based, consequence-based, combined). Evaluation of differences

between specific interventions would perhaps be more meaningful. However, limitations in the number of studies precluded such a granular evaluation and therefore were beyond the scope of this meta-analysis. Third, the Tau-U ES was the only ES calculated for this study. While Tau-U was chosen due to its advantages (e.g., greater statistical power in comparison to other non-overlap indices, control of autocorrelation; Parker et al., 2011), there is yet to be a standard ES within the field, as evidenced by the previous meta-analyses utilizing varying ES measures (de Bruin et al., 2013; Didden et al., 2006; Walker et al., 2018). Therefore, any interpretation of the reported effect sizes should be treated with caution. Fourth, this meta-analysis did not assess or control for publication bias. There was on overall lack of differential effects observed in this meta-analysis. Although there are several reasons, as previously noted, that could explain this finding, this lack of difference could be a function of publication bias. Previous studies have shown that studies with visually detectable effects are more likely to be published (Gage et al., 2017) and thus leading to generally positive effects and potentially making it difficult to detect any significant differences. Lastly, the generalizability of the findings is impacted due to the limited number of studies included ($n=30$) and the total number of participants ($n=45$). The number of participants is consistent with other meta-analyses reporting considerably low numbers for the adolescent age group (Bruni et al., 2017; Walker et al., 2018) compared to elementary and pre-school age groups. While the overall evidence base of behavior analytic interventions is strong, its applications towards adolescents with disabilities in school-based settings remain largely limited (David et al., 2021; Rivera et al., 2019).

Implications for Future Research and Practice

Several implications for future research were identified from the results of this meta-analysis. First, future research should include various measures of social validity. Only few studies ($n=9$) reported social validity and thus, this meta-analysis could not further examine its differences in effects. Information regarding social validity is important when considering the value of the intervention within the applied context. It is plausible that when typical agents find value in the intervention, they are likely to implement the intervention with higher fidelity, which in turn can lead to stronger effects. Additionally, considering the context and age group in which these interventions are implemented, it is important to determine whether these interventions are feasible enough to promote continued use. An intervention may be effective in reducing challenging behavior, but due to the complexity of the intervention or some other reasons (e.g., limited resource, difficulty to implement with older students), the relevant stakeholders (e.g., teachers and paraprofessional) may find little value in implementing the intervention. Furthermore, while there are no data to support it, implementing interventions in the naturalized environment may positively impact treatment maintenance and generalization.

Second, additional research is needed to determine whether or not interventions implemented in natural settings lead to improved maintenance and generalization. As noted previously, the intervention setting significantly moderated the effects of

challenging behavior interventions, indicating that the effects were more likely to be positive when implemented in naturalized contexts. However, to the authors' knowledge, there are no prior studies that examined the relationship between implementation setting and generalization and maintenance of effects. If an increase in maintenance and generalization is observed, this could further validate the intervention effects. Furthermore, if practitioners were found to continue utilizing the interventions, that could be a more accurate measure of social validity. Third, more high-quality research is needed for participants with other diagnoses, such as Down syndrome and fragile x syndrome. Although this study did not intend to limit the disabilities identified, the majority of the participants included had a primary disability of either ASD or ID. While the majority of students are likely to be identified as having either ASD or ID within school settings (National Center for Education Statistics, 2019), it is important that research is extended to those with lesser known diagnoses.

Third, future research should examine decision-making models for practitioners to determine when an experimental FBA is warranted. Although there are mixed findings on whether or not FBA is necessary and the results of this meta-analysis indicated that the type of FBA did not moderate the effects of the intervention, it is important to note that there are conditions that require FBA, particularly experimental FA (Lloyd et al., 2020). This highlights the importance of developing decision-making models for practitioners. Fourth, researchers should consider conducting in-depth analyses of ES measures given that previous research has shown that the ES measure used may influence the outcomes of the analysis (Walker et al., 2018). Furthermore, statistical analyses typically utilized in group research are now being expanded into single-case research (e.g., Heyvaert et al., 2012; Richman et al., 2015; Van den Noortgate & Onghena, 2008). Additional research is necessary to ascertain how these measures compare with the ES measures utilized in single-case research.

The results of this meta-analysis also yielded several implications for practice. First, findings of this meta-analysis support prior research indicating that there are no significant differences in treatment effects across individuals with DD. This finding is particularly encouraging for educators, as they are likely to serve individuals with varying disabilities. Second, as previously mentioned, the lack of significant differences observed between intervention outcomes based on whether a descriptive, experimental, or no FBA was conducted prior to intervention implementation suggests that conducting descriptive FBA might be sufficient in developing function-based interventions. This finding is encouraging given that FBA is a resource-intensive process, and specific FBA assessments (experimental FA) often require additional training or supervision from licensed professionals. Limited resources within schools tend to be a barrier for implementing FBAs in school settings. However, practitioners should be cautioned in determining whether or not an FBA is warranted as there are potential risks involved in implementing nonfunction-based interventions.

Adolescence is a sensitive period that may lead to increase in challenging behavior for individuals with DD. Although challenging behavior is one of the factors attributed to negative post-school outcomes, it is important that these behaviors are addressed prior to transitioning out of the school system, as services are limited for adults with DD (Manente, 2010). This decrease in available services highlights the

importance of addressing the needs of adolescents while they are still supported by the P-12 system. Outside the public education system, the provision of function-based treatment to adults with DD is not supported by any legal mandates. This meta-analysis provides emerging evidence for the efficacy of challenging behavior interventions in school settings for adolescents with DD and can guide educators in selecting effective challenging behavior interventions.

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Compliance with Ethical Standards

Informed Consent This study did not involve direct contact with any human or animal participants.

Conflicts of Interest All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.

Research Involving Human Participants and/or Animals This is a systematic review of the literature with quantitative measures of effect.

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